

Lab 3 - Routing Protocol

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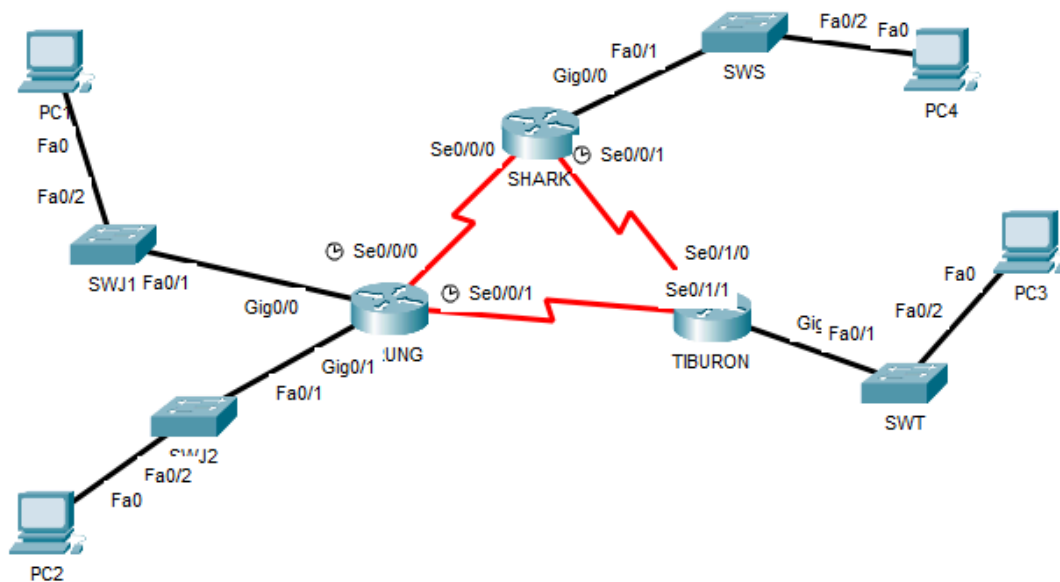
MATRIC NUM : A23CS0176

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.161	255.255.255.224	-
4		G0/1	172.18.110.129	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.190	255.255.255.224	172.18.110.161
12	PC2	-	172.18.110.158	255.255.255.224	172.18.110.129
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

SHARK:

- 60 host

$$- 2^x = 60$$

$$x = 5.91$$

$$\approx 6$$

$$2^6 = 64$$

172	18	110		
network position			Subnet	Host position
1010 1100 0001 0010 0110 111	000	000 000		
1010 1100 0001 0010 0110 111	000	111 111		

$$\begin{aligned} \text{- network portion} &= 32 - 6 \\ &= 26 \end{aligned}$$

- 62 usable IP address,
64 total IP address.

- Usable range:

$$172.18.110.1/26 - 172.18.110.62/26$$

$$\text{- network address: } 172.18.110.0/26$$

$$\text{- broadcast address: } 172.18.110.63/26$$

TIGERON:

- 50 host

$$- 2^x = 50$$

$$x = 5.91$$

$$\approx 6$$

$$2^6 = 64$$

172	18	110		
network position			Subnet	Host position
1010 1100 0001 0010 0110 111	001	000 000		
1010 1100 0001 0010 0110 111	001	111 111		

$$\begin{aligned} \text{- network portion} &= 32 - 6 \\ &= 26 \end{aligned}$$

- 62 usable IP address,
64 total IP address

- Usable range:

$$172.18.110.65/26 - 172.18.110.126/26$$

$$\text{- network address: } 172.18.110.64/26$$

$$\text{- broadcast address: } 172.18.110.127/26$$

JERUNG (LAN2):

- 30 host

$$- 2^n = 30$$

$$n = 4.91$$

$$\approx 5$$

$$2^5 = 32$$

172 18 110

Network position

Subnet Host
portion

1010 1100 0001 0010 0110 111 0 100 00000

1010 1100 0001 0010 0110 111 0 100 11111

$$\begin{aligned} \text{- network portion} &= 32 - 5 \\ &= 27 \end{aligned}$$

- 30 usable IP address,
32 total IP address.

- usable range:

172.18.110.129/27 - 172.18.110.158/27

- network address: 172.18.110.128/27

- broadcast address: 172.18.110.159/27

JERUNG (LAN1):

- 20 host

$$- 2^n = 20$$

$$n = 4.32$$

$$\approx 5$$

$$2^5 = 32$$

172 18 110

Network position

Subnet Host
portion

1010 1100 0001 0010 0110 111 0 101 00000

1010 1100 0001 0010 0110 111 0 101 11111

$$\begin{aligned} \text{- network portion} &= 32 - 5 \\ &= 27 \end{aligned}$$

- 30 usable IP address,
32 total IP address

- usable range:

172.18.110.161/27 - 172.18.110.190/27

- network address: 172.18.110.160/27

- broadcast address: 172.18.110.191/27

JERUNG-SHARK

- 2 host

$$- 2^x = 2$$

$$x = 1$$

$$2^x = 2$$

172 18 110

Network position

Subnet

Host
Portion

10101100 00010010 0110111

0110000

00

10101100 00010010 0110111

0110000

11

$$\begin{aligned} \text{- network portion} &= 32 - 2 \\ &= 30 \end{aligned}$$

- usable range:

172.18.110.193/30 - 172.18.110.194/30

- network address: 172.18.110.192/30

- broadcast address: 172.18.110.195/30

JERUNG-LIBURON:

- 2 host

172 18 110

Network position

Subnet

Host
portion

$$- 2^x = 2$$

10101100 00010010 0110111

0110001

00

$$x = 1$$

10101100 00010010 0110111

0110001

11

$$2^x = 2$$

$$\begin{aligned} \text{network portion} &= 32 - 2 \\ &= 30 \end{aligned}$$

- usable range:

172.18.110.197/30 - 172.18.110.198/30

- network address: 172.18.110.196/30

- broadcast address: 172.18.110.199/30

SHARK - TIGERON				
- 2 hosts				
	172	18	110	
	Network position			Subnet
$-2^x = 2$				Host position
$x = 1$	1010	1100	0001 0010 0110 111	011 0010
$2^1 = 2$	1010	1100	0001 0010 0110 111	011 0010
- network position: $32 - 2$				
$= 30$				
- usable range:				
172.18.110.201/30 - 172.18.110.202/30				
- network address: 172.18.110.200/30				
- broadcast address: 172.18.110.203/30				
PC1:				
IP address: 172.18.110.190/27				
gateway: 172.18.110.161/27				
PC2:				
IP address: 172.18.110.158/27				
gateway: 172.18.110.129/27				
PC3: 172.18.110.126/26 (IP address)				
gateway: 172.18.110.65/26				
PC4				
IP address: 172.18.110.62/26				
gateway: 172.18.110.1/26				

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to do this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

```
JERUNG>enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/1
L       172.18.110.129/32 is directly connected, GigabitEthernet0/1
C       172.18.110.160/27 is directly connected, GigabitEthernet0/0
L       172.18.110.161/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
```

Figure 1: JERUNG


```

SHARK>enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

Figure 2: SHARK

```

TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#

```

Figure 3: TIBURON

b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

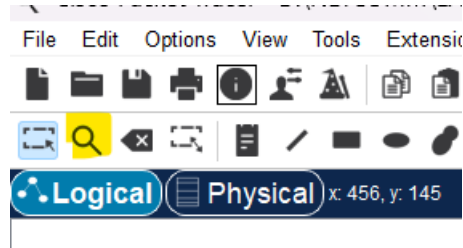


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Figure 4: JERUNG

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Figure 5: SHARK

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Figure 6: TIBURON

2. Try to ping PC2 from PC1, paste the results here.

```
Pinging 172.18.110.158 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.158: bytes=32 time<1ms TTL=127
Reply from 172.18.110.158: bytes=32 time=10ms TTL=127
Reply from 172.18.110.158: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

PC1 to PC2 successfully ping because both PCs are connected to the same router. PC1 cannot ping PC4 because PC1 and PC4 are both connected to different routers which are not interconnected. The packets shared between the subnets are not forwarded as there is no way into the subnet.

5. What needs to be done to ensure all PCs can ping each other successfully?

By making sure the IP address and subnet masks for all PCs are correct and match to the network's configuration.

Task 3 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Figure 7: TIBURON and JERUNG before RIP

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the ‘network’ instructions in Figure E?

The network instructions apply subnet addresses to determine which networks that RIP (Routing Internet Protocol) should advertise and include in its routing updates. The ‘no auto-summary’ command disables classful analysis of network boundaries, in order to maintain more specific routing information.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Figure 8: TIBURON and JERUNG after RIP

- a. What are the changes seen in TIBURON?

The “Next Hop IP” refers to the IP address of the corresponding router, with a metric of 120/1. This signifies that the destination is reachable within one hop, indicating direct connectivity through a single router.

- b. What are the Networks with type R in TIBURON and JERUNG?

In TIBURON :

- 172.18.110.128/27
- 172.18.110.192/30
- 172.18.110.160/27

In JERUNG :

- 172.18.110.64/26
- 172.18.110.200/30

- c. Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=20ms TTL=126
Reply from 172.18.110.126: bytes=32 time=25ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 15ms
```

Yes, it is successful.

- d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.
Reply from 172.18.110.161: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No, it was unsuccessful.

- e. Explain the reasons for your answer.

The routers of PC1 and PC4, which are SHARK and JERUNG respectively, are not connected. Therefore packets to be sent dropped along the way. While PC1 and PC3 successfully ping because both PCs connected to the same router.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=20ms TTL=126 Reply from 172.18.110.190: bytes=32 time=16ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 20ms, Average = 9ms C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=12ms TTL=126 Reply from 172.18.110.190: bytes=32 time=6ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 5ms </pre>
2	<pre> C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=10ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=11ms TTL=126 Reply from 172.18.110.158: bytes=32 time=16ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 16ms, Average = 9ms C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=15ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=3ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 15ms, Average = 5ms </pre>

3	<pre>C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=21ms TTL=126 Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 21ms, Average = 6ms C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=23ms TTL=126 Reply from 172.18.110.126: bytes=32 time=9ms TTL=126 Reply from 172.18.110.126: bytes=32 time=8ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 23ms, Average = 10ms</pre>
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Task 4 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: **192.168.1.0**
 - b. As this change happens, PC3 must also have a different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.3
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?

No.

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
L	172.18.110.194/32	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
					L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.

From the routing table, SHARK lacks an entry for the 192.168.1.0/24 network, while TIBURON has the entry on GigabitEthernet0/0. The absence of this route means SHARK cannot forward packets to the 192.168.1.0/24 subnet. This missing route highlights the gap in SHARK's routing configuration, which could result in end-to-end connectivity issues.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

To maintain end-to-end connectivity, update the PC configurations with the correct gateway and IP addresses. Additionally, verify connectivity by pinging all devices from each PC to ensure proper connectivity.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre>C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=3ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 3ms, Average = 1ms</pre>
2	<pre>C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=22ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=9ms TTL=126 Reply from 172.18.110.158: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 22ms, Average = 11ms</pre>

4	<pre>C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=6ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 6ms, Average = 2ms</pre>
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REFLECTION

From this lab, I gained a clear understanding of subnetting and how to calculate and configure IP addresses for different devices. Setting up the routing tables and working with RIP protocol allowed me to see how routers communicate and exchange routing information, I also learned how to identify and fix connectivity issues, such as missing routes, which can cause network interruptions. Troubleshooting with the ping test helped me understand the importance of verifying configurations to maintain proper connections. Overall, this lab 3 strengthened my skills in configuring and analyzing network setups.